

ELVIS TANG

Embedded Systems | Firmware Engineering | C/C++

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Computer Engineering graduate with hands-on experience developing embedded C/C++ software for sensor-fusion and autonomous vehicle systems. Experienced with UART, SPI, I2C, Raspberry Pi, Arduino, real-time data acquisition, hardware/software integration, and system testing.

EXPERIENCE

Embedded Systems Project Lead - Lockheed Martin-Sponsored Capstone

Aug 2022 - May 2023

California State Polytechnic University, Pomona

Pomona, CA

- Developed and debugged embedded C/C++ software for an autonomous ground vehicle integrating LiDAR, stereo camera, ultrasonic sensors, IMU, and wheel encoders on Raspberry Pi and Arduino platforms.
- Implemented and tested UART, SPI, and I2C communication and Kalman-filter-based sensor fusion for real-time data acquisition and obstacle avoidance.
- Validated distance measurements within +/-10 cm on varied surfaces and demonstrated real-time object classification up to 50 cm during navigation trials.

Instructor

Nov 2023 - Sep 2024

Coding Minds Academy

Irvine, CA

- Taught Python, JavaScript, Git, game development, and introductory hardware concepts to classes of 10+ students.
- Guided student software and hardware projects through design, implementation, debugging, and demonstration.
- Developed project-based curricula that connected programming fundamentals with game development and real-world engineering tools.

Stand Lead

Aug 2024 - Present

Aramark

Anaheim, CA

- Patch and troubleshoot point-of-sale systems to keep registers operational during high-volume events.
- Coordinate daily concession operations serving 1,000+ guests per shift and resolve 20+ customer or operational issues.
- Manage inventory for 50+ items while maintaining food-safety, cash-handling, and workplace compliance procedures.

SELECTED PROJECTS

Sensor Fusion and Tracking for an Autonomous Ground Vehicle | Lockheed Martin, C/C++, Raspberry Pi, Arduino

- Integrated LiDAR, stereo camera, ultrasonic sensors, IMU, and wheel encoders through complementary and Kalman-filter-based sensor fusion.
- Validated obstacle avoidance and navigation across varied surfaces, including distance measurements within +/-10 cm and object classification up to 50 cm.
- Implemented UART, SPI, and I2C interfaces between Raspberry Pi, Arduino, sensors, and controllers to support real-time data acquisition.

Open-Source Haptic Virtual Reality Gloves | Embedded Hardware, Haptics, Prototyping

- Built a wearable hardware prototype exploring tactile feedback and physical interaction for virtual-reality input systems.
- Designed sensing, actuation, and feedback-loop concepts that translate hand movement into responsive user-facing VR behavior.
- Documented the open-source prototype and evaluated embedded design choices affecting comfort, responsiveness, and interaction.

Comparing x86 and AMD Processor Performance | SilverBench, Benchmarking, Computer Architecture

- Ran SilverBench workloads to compare processor performance using a repeatable benchmarking methodology and observed performance signals.
- Analyzed benchmark results in relation to processor architecture, computational throughput, and system-level performance tradeoffs.
- Presented the test setup, observations, and architecture-level conclusions in a structured technical comparison.

EDUCATION & CERTIFICATIONS

B.S., Computer Engineering - California State Polytechnic University, Pomona

Additional Certification: CompTIA Security+ (SY0-701)

TECHNICAL SKILLS

Firmware: C/C++, Python, embedded debugging, real-time data acquisition, UART, SPI, I2C

Platforms & Hardware: Raspberry Pi, Arduino, LiDAR, stereo camera, ultrasonic sensors, IMU, wheel encoders

Methods & Tools: sensor fusion, Kalman filtering, obstacle avoidance, Git, GitHub, MATLAB